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Comparison of proposed methods to include lightweight framed structures in EN 12354 prediction model

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Summary

EN 12354 gives a well-established framework for the prediction of the acoustic system performance of buildings using input data of the performance of buildings elements, like transmission loss, damping and junction coupling. Originally, the method was developed for homogenous monolithic building elements with a high-modal density, small attenuation of structure-borne waves and a low coincidence frequency, like concrete and masonry structures. However, often lightweight framed elements, e.g. wood or steel stud walls and joist floors, are used in modern buildings that do not fulfill most of these conditions. Nevertheless, an inclusion of these structures in the EN 12354 framework would be desirable for building designers and in the recent COST FP0702 Action different methods to include those elements were proposed. In this paper proposed methods are compared through prediction of flanking sound transmission across a wood frame wall-wall junction. Data measured in the NRC-IRC flanking sound transmission facility is used as input data as well as for the validation of the methods.

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1. Introduction

EN 12354 gives a well-established framework for the prediction of sound transmission in buildings, of the so-called system performance. It takes into account direct sound transmission through the separating building element between two adjoining spaces as well as flanking sound transmission that involves other building elements than the partition, e.g. sidewalls that are coupled at a building junction to the partition wall. Predictions are performed with measured and/or predicted input data of the element performance and of structure-borne sound transmission across the junction. Originally, the methods were developed for “heavy” homogenous monolithic structures that are weakly damped, like concrete or masonry. However, often lightweight framed elements, like framed walls and joist floors, are used in modern buildings that do not fulfill most of these conditions. One goal of COST FP0702-Action was to develop the necessary knowledge to include lightweight framed elements in the EN 12354 framework and different methods were proposed. This paper only focuses on the prediction airborne sound transmission; nevertheless, the used

approaches can also easily be adjusted for impact sound.

2. Methods to predict airborne sound transmission in lightweight framed buildings

Sound transmission through lightweight framed double leaf assemblies is more complex than for single leaf walls and floor. For instance, the structure-borne sound velocity can differ greatly on both surfaces of a double leaf element due to the sound attenuation by the frame and cavities, whereas on both surfaces of a homogenous monolithic element they are assumed to be equal. Further, the coincidence frequency of the leaves of framed elements is usually very high and in the frequency range below it must be distinguished between resonant and non-resonant transmission. Both transmission components are related to structure-borne sound velocity components, the resonant due to free bending waves and the non-resonant due to forced bending waves on the leaves.

As for monolithic elements, it is also assumed that only the resonant structure-borne velocity component is transmitted structurally from one element to the other across a building junction.

Finally, the assumption of a diffuse structure-borne sound field with an statistically even distribution of velocity on one element also does not hold for most lightweight framed assemblies, because structure-borne sound waves are much more attenuated than on homogenous monolithic elements.

However, it should be kept in mind that EN 12354 is an engineering prediction method and an accurate prediction method for the required input data that takes into account all these aspects is not readily available yet. Therefore, an approach that uses measured input data would be most desirable for EN 12354. In the past several approaches for the prediction of airborne sound transmission were proposed that are all very similar and follow the basic EN 12354 equations for monolithic homogenous construction. In the following three approaches that are evaluated in this paper are outlined and discussed for the case of flanking sound transmission across a wall-wall junction.

2.1. CSTB-Approach

CSTB presented their approach to measure and predict flanking sound transmission in two companion papers [1], [2]. If direction averaging is applied, the flanking sound reduction index R_{ij} through the two coupled building elements i and j is given by Equation 1.

$$R_{ij,CSTB} = \frac{R_i + R_j}{2} + \overline{D_{va,ij}} + 5 \lg \frac{\sigma_{a,i} \sigma_{a,j}}{\sigma_{s,i} \sigma_{s,j}} + 5 \lg \frac{S_s^2}{S_j S_i}, \quad (1)$$

The sound reduction indices R_i and R_j of the elements measured according to ISO 10140 are used that include both resonant and non-resonant transmission components.

$$\overline{D_{va,ij}} = D_{va,Ij} + D_{va,Ji} \quad (2)$$

Further, the direction averaged velocity level difference $D_{va,ij}$ is necessary as defined in Equation 2. The velocity level difference $D_{va,Ij}$ is measured between the non-exposed leaf I of the element on the source side and leaf j on the receiving side that radiates sound in the receive room, while room i is excited with airborne sound, as depicted in Figure 1 for the flanking element-flanking element path. Accordingly, $D_{va,Ji}$ is measured for the opposite direction.

The radiation efficiencies $\sigma_{a,I}$ and $\sigma_{a,J}$ of leaves I and J are measured when room i and j respectively are excited with airborne sound. These radiation efficiencies capture resonant as well as non-resonant transmission. The radiation efficiencies $\sigma_{s,i}$ and $\sigma_{s,j}$ of leaf i and of leaf j

respectively are for resonant radiation only and are measured while the radiating element is excited with a structural source, like a shaker or impact hammer. The last term contains the surface area of the separating element S_s as well as of the flanking elements S_i and S_j .

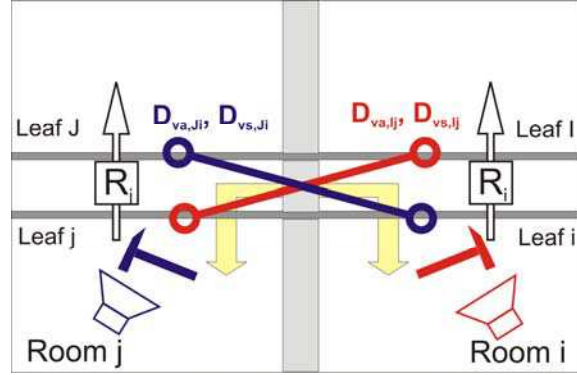


Figure 1. Element denotations and measurement positions

2.2. Full - Approach

In another study terms for the prediction of flanking transmission were derived following a statistical energy analysis (SEA) scheme, but using a more empirical way that is called modified SEA approach in [3]. A term given by Equation 3 was found that is quite similar to the CSTB approach.

$$R_{ij,full} = \frac{R_{r,i} + R_{r,j}}{2} + \overline{D_{vs,ij}} + 5 \lg \frac{\sigma_{s,j} \sigma_{s,i}}{\sigma_{s,i} \sigma_{s,j}} + 5 \lg \frac{S_s^2}{S_j S_i}, \quad (3)$$

$$\overline{D_{vs,ij}} = D_{vs,Ij} + D_{vs,Ji}, \quad (4)$$

Below the coincidence frequency only the resonant components $R_{r,i}$ and $R_{r,j}$ of the sound reduction indices are used as input data. In [3] these quantities were predicted using SEA, but for the matter of simplicity estimates as described later in section 3.1 were used in this paper.

The velocity level differences in Equation 4 are measured between the same leaves as in the CSTB approach, but the leaves in room i and j respectively are excited with a structure-borne source instead of using airborne sound. Similarly, the radiation efficiencies $\sigma_{s,i}$ and $\sigma_{s,j}$ of leaf I and leaf J respectively are for resonant radiation only and are measured while the radiating element is excited structurally.

2.3. EN 12354 Approach

Gerretsen [4] proposed a method that is similar to the full approach above and following closer the equations of EN 12354 for monolithic construction.

$$R_{ij,EN 12354} = \frac{R_{r,i} + R_{r,j}}{2} + \overline{D_{vs,ij}} + 5 \lg \frac{S_s^2}{S_i S_j}, \quad (5)$$

Equation 5 is identical to Equation 3 except that the third term with the radiation efficiencies that relates the transmitted sound power to the surface velocities is neglected in Equation 5. It is assumed that the radiation efficiencies of both leaves of a building element (e.g. $\sigma_{s,i}$ and $\sigma_{s,j}$) are similar and hence cancel out which may hold for many situations.

3. Input Data for Prediction

Most of the input data used in this paper will be either measured or derived from measured data.

3.1. Resonant Sound Reduction Index

As proposed by Villot in [2], the resonant component of the sound reduction index $R_{r,i}$ will be estimated from the sound reduction index R_i measured according to ISO 10140. Further, the ratio of the radiation efficiency $\sigma_{a,i}$ for airborne excitation in room i and $\sigma_{s,i}$ for structure-borne excitation of element i measured at the non-exposed surface of the flanking element.

$$R_{r,i} = R_i + C_{r,i} = R_i + 10 \lg \frac{\sigma_{a,i}}{\sigma_{s,i}}, \quad (6)$$

The direct sound reduction indices used in this paper were measured according to ISO 10848 as described in Section 4 at a flanking test specimen that was installed in the NRC-Construction Flanking Facility.

3.2. Radiation efficiency

The radiation efficiency in Equation 7 is defined as ratio of sound power $W_{rad,i}$ that is actually radiated by a surface S_i into a space and the sound power that would be theoretically radiated by an ideal piston that moves with the same space average mean square surface velocity $\langle v_i \rangle^2$ as the radiating surface where ρ_0 and c_0 are the density and sound speed in air.

$$\sigma_i = \frac{W_{rad,i}}{\rho_0 c_0 S_i \langle v_i \rangle^2}, \quad (7)$$

Unfortunately, currently no standard or guideline exists for the measurement of the radiation efficiency itself. Nevertheless, in this paper parts of the ISO 10140 protocol were applied for the sound power measurement and parts of the ISO 10848 protocol for the surface velocity.

3.3. Velocity Level Differences

ISO 10848 gives with the so-called direct method a protocol for the measurement of the velocity level differences at building junctions. However, ISO 10848 was originally developed for homogenous, monolithic building elements. For lightweight double leaf elements care has to be taken that the right side of the element is excited and also velocity levels are measured on the appropriate leaves as defined in Section 2 depending on the selected approach.

4. Measurement of Flanking Sound Reduction Index

As reference to gauge the accuracy of the different prediction approaches, the flanking sound reduction index R_{ff} was directly measured using the so-called indirect method of ISO 10848. Hereby, a full scale wood frame junction specimen consisting of 8 walls and 4 floors was installed in the NRC-Construction Flanking Facility. The elements divide the space inside the outer shell of the facility in eight rooms with four located on each of the two floors. Flanking sound transmission between the rooms through the structure of the facility is suppressed so that only transmission through the specimen can be measured. The rooms were excited with airborne noise and sound pressure levels were measured at nine microphone positions in all eight rooms. The facility is equipped with an automated measurement system that positions the microphone in each room exactly at the same locations to ensure a small uncertainty for the measurement repeatability. In order to extract all relevant path data as described in more detail in [5] sound transmission paths have to be sequentially suppressed during measurements. To do so additional layers of gypsum board on a layer of fibrous insulation material were placed without rigid connections to the test specimen in front of the element surfaces.

5. Flanking Test Specimen

As flanking test specimen typical North American wood frame construction consisting of wood framed walls and wood I-joist floors was installed in the flanking facility. All presented measurements are conducted at the wall-wall cross

junction in the lower rooms with junction details shown in Figure 2. The walls under test consisted of two layers of 16 mm gypsum board directly attached to a wood framing (38 x 89 mm) with studs spaced 400 mm on centre on one side and a single layer of 16 mm gypsum board mounted on resilient channels spaced at 600 mm on the second side. The wall cavities were filled with 90 mm glass fiber insulation. The surfaces with the directly attached gypsum board are facing the source and receiving room of this study. At the junction the gypsum board was discontinuous whereas the footer and header of the test wall framing were continuous.

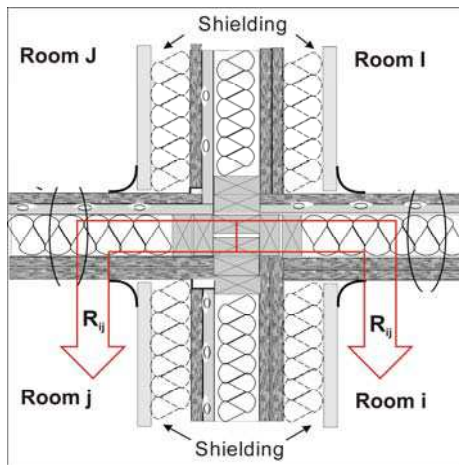


Figure 2. Wall-wall junction under test

6. Results

The velocity levels for the velocity level differences and radiation efficiencies presented below were measured on the whole wall surfaces in point grids with mesh sizes between 20 cm and 30 cm using a scanning laser vibrometer system. The sound pressure levels to calculate the radiation efficiency were measured for each excitation position at the same nine microphone positions as for the sound reduction measurements. For structure-borne excitation each element was excited at 7 points with an electro-dynamic shaker. For airborne excitation the installed sound system in the facility was used, and shielding was applied to other surfaces of the specimen to suppress excitation of other surfaces than the one under test. Unfortunately, the combination of used shaker and velocity measurement system was only optimal for velocity level difference measurements in the frequency range including the 125 Hz to 2000 Hz

octave bands. At frequencies below the shaker was not powerful enough to excite the system efficiently and above the velocity measurements on the receive side were affected by the limited dynamic range of the measurement system.

6.1. Radiation Efficiency

There was a good agreement between the radiation efficiencies for similar leaves of both walls. Thus, for the matter of clarity, only the average radiation indices for the two coupled walls are presented for the structure-borne and airborne excitation case in Figure 3.

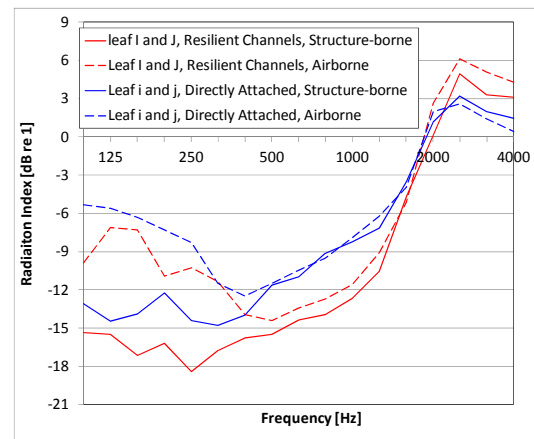


Figure 3. Measured radiation index of gypsum board leaves for airborne and structure-borne excitation

The radiation index of the gypsum board leaf that is mounted on resilient channels is for both airborne and structure-borne excitation about 3 dB lower than for the directly attached leaf in most of frequency range below coincidence. This is due to the weaker restraint of the gypsum board when mounted to resilient channels and the lesser inhomogeneities in the structure-borne sound field that enhance sound radiation below coincidence. For both attachment methods the radiation index for airborne excitation are almost equal to structure-borne excitation in the mid-frequency range. Only towards the low frequency range, the airborne radiation index increases. This is in good agreement with what will be expected for double leaf wood frame walls with cavity insulation. Forced or non-resonant transmission of airborne waves through the cavity is well attenuated in the mid and high frequency range by the cavity absorption whereas structure-borne and hence resonant sound transmission due to the coupling of the leaves by the wooden frame dominates. At low frequencies, the cavity absorption is less effective

and wall sound insulation is further compromised by the mass-spring-mass resonances of the leaves that also belong to forced transmission regime.

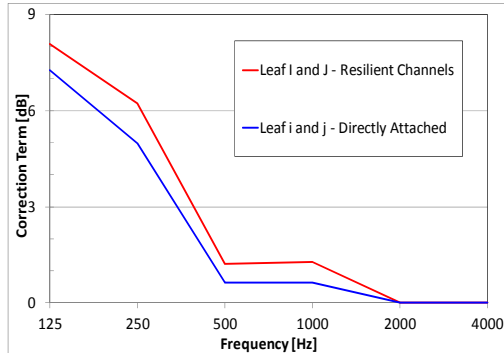


Figure 4. Correction term C of Equation 6 for non-resonant transmission

The correction factor C of Equation 6 for resonant sound transmission which is basically the difference of the radiation index for airborne and structure-borne excitation of Figure 3 is presented for both gypsum board attachment methods in octave bands in Figure 4. Because of reciprocity in theory the difference should be equal for both wall surfaces; however, the difference is about 1 dB which gives an estimate of the uncertainty for the radiation efficiency measurements. The correction is with about 7-8 dB biggest at 125 Hz, approaches 0 dB well below coincidence and is set manually to 0 dB above coincidence where only resonant transmission occurs. The correction cannot be applied to other double leaf assemblies and can be expected to be much greater if e.g. the cavity would be less absorbent and/or structural coupling between the leaves weaker.

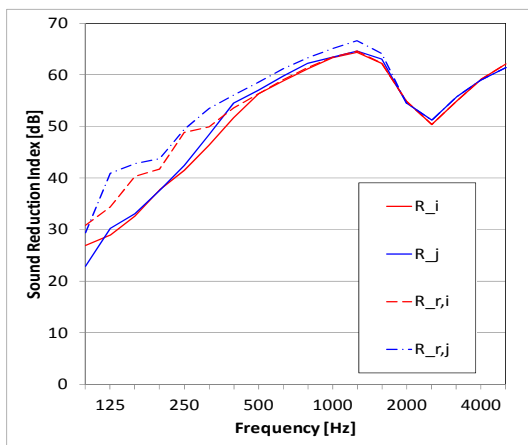


Figure 5. Measured sound reduction index R and resonant sound reduction R_r of both wall assemblies

6.2. Sound Reduction Index

In Figure 5 the direct sound reduction index for the two walls are shown as measured in the test facility and as resonant component R_r with the correction term C applied. Again the differences between both coupled walls are very small with 3 dB at most in one third octave band.

6.3. Velocity Level Differences

The measured velocity level differences are presented for the two excitation cases in Figure 6 in octave bands to show better global trends. In the low frequency range the velocity level differences are for airborne excitation about 3 to 4 dB greater than for structure-borne because of the non-resonant velocity component. In the mid-frequency range the velocity level differences are almost equal and spread in a range of 2 dB.

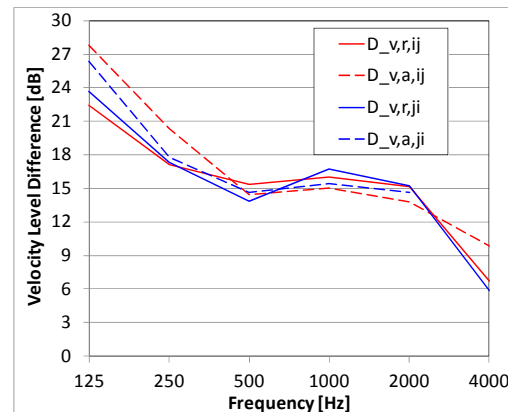


Figure 6. Measured velocity level differences with structureborne and airborne excitation

The difference between the two excitation cases is less for the measured velocities than for the radiation index because below coincidence a small non-resonant velocity component radiates sound much more efficiently than the resonant. This is also reflected by the velocity level differences.

6.4. Flanking Sound Reduction Index

The predicted and measured flanking sound reduction indices are presented in Figure 7 for a single flanking path. All three prediction approaches agree well with the measurement, but the EN 12354 approach tends to slightly overestimate the flanking sound reduction index in the mid frequency range. The great difference between prediction and measurement above 2000 Hz is due to problems with measurement of

the surface velocity on the receive leaf as already discussed earlier.

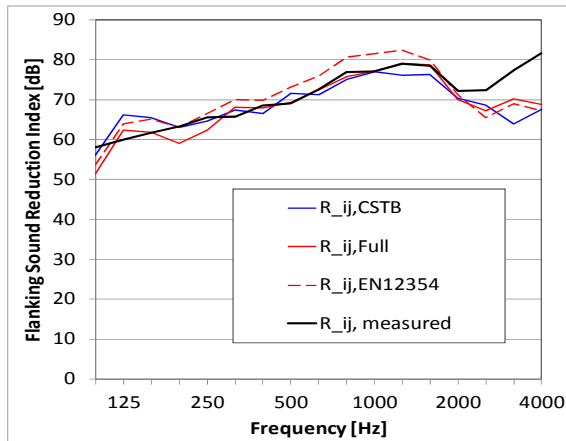


Figure 7. Measured and predicted flanking sound reduction index R_{ij}

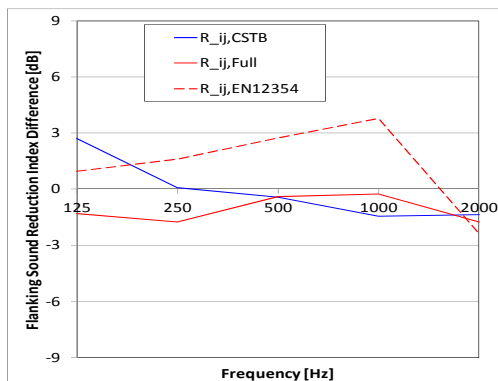


Figure 8. Difference between measured and predicted flanking sound reduction index R_{ij}

The difference between prediction and measurement is shown in Figure 8 in octave bands to show the more global trend. All differences are about in a range of about ± 3 dB at most. The difference between the EN 12354 approach and the full approach is about 3 dB below coincidence and agrees very well with the difference in the radiation efficiencies for the gypsum board mountings that were assumed to be zero in the EN 12354 approach. The difference between the CSTB approach and full approach is very small in the mid frequency range and increase to about 3 dB in the low frequency range.

7. Conclusions

For the junction under test it was found that all three methods are suitable for the prediction of the flanking sound reduction index.

Special care is necessary since the application of

the EN 12354 approach is limited to cases where the radiation efficiencies are equal or very similar on both sides of the coupled double leaf elements. That this may not always be the case was shown in the example where gypsum board on one side of the wall was directly attached to the wood frame and on the other mounted on resilient channels. However, the error in prediction was only 3 dB, but certainly cases exist, e.g. hybrid construction with a thin concrete slab floor on steel joists and gypsum board at the walls, where the error can be expected to be much bigger.

Both CSTB approach and full approach predict the flanking sound reduction index for the current case well. Inspection of Equation 5 and Equation 6 reveals that the CSTB and the full approach are actually identical except in the measurement of the velocity level differences, where first uses airborne excitation and latter structure-borne excitation. The resulting difference in both was small for the current junction. It is suggested to gauge the suitability of both approaches on another assembly, e.g. one without cavity insulation, where the non-resonant velocity component is expected to be much greater. The need for airborne excitation in the velocity measurement is more demanding, as shielding is required to avoid transmission through non-considered flanking paths.

Generally, also guidelines for the radiation efficiency measurement are desirable and necessary if either of the three approaches is included in EN 12354.

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